

LIBRARY SEAT RESERVATION SYSTEM

PROGRAM:

```
import java.util.Scanner;

public class LibrarySeatReservation {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int totalSeats = 10;
        boolean[] seats = new boolean[totalSeats];

        int choice;

        do {
            System.out.println("\n===== LIBRARY SEAT RESERVATION =====");
            System.out.println("1. View Seats");
            System.out.println("2. Reserve Seat");
            System.out.println("3. Cancel Reservation");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");

            choice = sc.nextInt();

            switch (choice) {

                case 1:
                    System.out.println("\nSeat Status:");

                    for (int i = 0; i < totalSeats; i++) {
                        if (seats[i] == false)
                            System.out.println("Seat " + (i + 1) + " - Available");
                        else
                            System.out.println("Seat " + (i + 1) + " - Reserved");
                    }
                    break;

                case 2:
                    System.out.print("Enter seat number to reserve: ");
                    int reserve = sc.nextInt();
```

```

        if (reserve < 1 || reserve > totalSeats) {
            System.out.println("Invalid seat number!");
        }
        else if (seats[reserve - 1]) {
            System.out.println("Seat already reserved!");
        }
        else {
            seats[reserve - 1] = true;
            System.out.println("Seat " + reserve + " reserved successfully!");
        }
        break;

    case 3:
        System.out.print("Enter seat number to cancel: ");
        int cancel = sc.nextInt();

        if (cancel < 1 || cancel > totalSeats) {
            System.out.println("Invalid seat number!");
        }
        else if (!seats[cancel - 1]) {
            System.out.println("Seat is not reserved!");
        }
        else {
            seats[cancel - 1] = false;
            System.out.println("Reservation cancelled successfully!");
        }
        break;

    case 4:
        System.out.println("Thank you!");
        break;

    default:
        System.out.println("Invalid choice!");
    }

} while (choice != 4);

sc.close();
}
}

```

OUTPUT:

===== LIBRARY SEAT RESERVATION =====

1. View Seats
2. Reserve Seat
3. Cancel Reservation
4. Exit

Enter your choice: 1

Seat Status:

Seat 1 - Available
Seat 2 - Available
Seat 3 - Available
Seat 4 - Available
Seat 5 - Available
Seat 6 - Available
Seat 7 - Available
Seat 8 - Available
Seat 9 - Available
Seat 10 - Available

Enter your choice: 2

Enter seat number to reserve: 5

Seat 5 reserved successfully!

Enter your choice: 1

Seat 5 - Reserved